

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-23280410})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary ideal I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{(1-\sigma_x)^2}(t) J \mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}3306 + \mathbf{Z}(922 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K . The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 23280410 = 0$, of discriminant -93121640 . Class group invariants $(90 \ 10 \ 10)$, class number 9000; the three stored generator ideals (HNF) are $\begin{pmatrix} 110 & 10 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2783 & 2243 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5313 & 3002 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1223)y^3 + (31s - 54702)y^2 + (-262s - 77374)y + (5362s + 13916100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2442y^8 - 104512y^7 + 24840199y^6 - 1281442632y^5 \\ & + 37697310058y^4 - 653399852384y^3 + 7821002817556y^2 - 67564177 \\ & 254880y + 862993931458040 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $92583348642112 = 2^6 \cdot 373 \cdot 3878323921$:

$$\begin{pmatrix} 2893229645066 & 676961525184 & 589502920126 & 52456742812 & 1664024087210 & 1604042756432 & 2768392970129 & 1370328784183 & 286764359040 & 318371394619 \\ 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 331 factors with signed exponents of absolute value up to 6678860368142069536232957063; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1223)y^3 + (31s - 54702)y^2 + (-262s - 77374)y + (5362s + 13916100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2442y^8 - 104512y^7 + 24840199y^6 - 1281442632y^5 \\ & + 37697310058y^4 - 653399852384y^3 + 7821002817556y^2 - 67564177 \\ & 254880y + 862993931458040 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2 = 2$:

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 335 factors with signed exponents of absolute value up to 8630016995894005890277435997; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1223)y^3 + (31s - 54702)y^2 + (-262s - 77374)y + (5362s + 13916100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2442y^8 - 104512y^7 + 24840199y^6 - 1281442632y^5 \\ & + 37697310058y^4 - 653399852384y^3 + 7821002817556y^2 - 67564177 \\ & 254880y + 862993931458040 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right) s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $70723915378000 = 2^4 \cdot 5^3 \cdot 83 \cdot 317 \cdot 349 \cdot 3851$:

$$\begin{pmatrix} 353619576890 & 226223787320 & 304845758910 & 184862879596 & 184632968166 & 181072105461 & 294212362154 & 138662390884 & 2477451635 & 239748746535 \\ 0 & 10 & 0 & 2 & 9 & 5 & 2 & 1 & 8 & 3 \\ 0 & 0 & 10 & 4 & 1 & 5 & 4 & 5 & 1 & 8 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 335 factors with signed exponents of absolute value up to 11360477318744101310774891592; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1386y^3 + (8s + 3762)y^2 + (596s - 3739335)y + (-540s + 9732000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2768y^8 + 13068y^7 - 5572722y^6 + 23993076y^5 + 1 \\ & 1830607504y^4 + 166890129220y^3 + 22124281186385y^2 - 87767550748 \\ & 800y + 101500391556000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944}\right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $8225582389200 = 2^4 \cdot 3^3 \cdot 5^2 \cdot 11 \cdot 17^4 \cdot 829$:

$$\begin{pmatrix} 79061730 & 46549740 & 35701836 & 56876842 & 57146203 & 68486330 & 39987348 & 59418310 & 11589448 & 11429765 \\ 0 & 510 & 306 & 496 & 93 & 36 & 410 & 103 & 230 & 31 \\ 0 & 0 & 102 & 0 & 66 & 44 & 0 & 38 & 14 & 86 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 144 factors with signed exponents of absolute value up to 1194421; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1386y^3 + (8s + 3762)y^2 + (596s - 3739335)y + (-540s + 9732000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2768y^8 + 13068y^7 - 5572722y^6 + 23993076y^5 + 1 \\ & 1830607504y^4 + 166890129220y^3 + 22124281186385y^2 - 87767550748 \\ & 800y + 101500391556000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25897180844700 = 2^2 \cdot 3^3 \cdot 5^2 \cdot 11 \cdot 17^2 \cdot 3017159$:

$$\begin{pmatrix} 50778785970 & 9994992240 & 46721424228 & 50392711342 & 21352507294 & 8267213598 & 901181427 & 47146733096 & 28986564526 & 2818175337 \\ 0 & 255 & 44 & 145 & 235 & 185 & 10 & 30 & 254 & 33 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 144 factors with signed exponents of absolute value up to 150820; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1386y^3 + (8s + 3762)y^2 + (596s - 3739335)y + (-540s + 9732000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2768y^8 + 13068y^7 - 5572722y^6 + 23993076y^5 + 1 \\ & 1830607504y^4 + 166890129220y^3 + 22124281186385y^2 - 87767550748 \\ & 800y + 101500391556000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $437679842688000 = 2^{10} \cdot 3^6 \cdot 5^3 \cdot 11 \cdot 653^2$:

$$\begin{pmatrix} 7757640 & 1175400 & 4513800 & 3608708 & 7472272 & 2541578 & 4206062 & 4256803 & 309334 & 5024847 \\ 0 & 117540 & 50040 & 36236 & 105080 & 63184 & 54798 & 53598 & 53436 & 57559 \\ 0 & 0 & 30 & 18 & 28 & 18 & 24 & 20 & 26 & 28 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 154 factors with signed exponents of absolute value up to 210093297; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1497y^3 - 6sy^2 + 2485196y + 5144s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2994y^8 + 7211401y^6 + 8278771584y^4 + 4739146009936y^2 + 616016782981760$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6315714281800 = 2^3 \cdot 5^2 \cdot 7^2 \cdot 11^2 \cdot 331 \cdot 16091$:

$$\begin{pmatrix} 4101113170 & 3067607340 & 1559075136 & 594349927 & 1415021108 & 1424640901 & 1723410576 & 3394723676 & 2276746911 & 3299080474 \\ 0 & 70 & 35 & 34 & 57 & 0 & 23 & 57 & 23 & 53 \\ 0 & 0 & 11 & 6 & 9 & 5 & 10 & 5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 191 factors with signed exponents of absolute value up to 136699532599658633497440783230; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1497y^3 - 6sy^2 + 2485196y + 5144s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2994y^8 + 7211401y^6 + 8278771584y^4 + 4739146009936y^2 + 616016782981760$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $205986867915136 = 2^7 \cdot 7 \cdot 11^3 \cdot 97 \cdot 1780663$:

$$\begin{pmatrix} 1170379931336 & 644981247460 & 433569542132 & 559686401357 & 397452442684 & 697936890398 & 236541726330 & 669833112681 & 1059099272518 & 355516631038 \\ 0 & 44 & 32 & 19 & 38 & 40 & 26 & 36 & 38 & 31 \\ 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 191 factors with signed exponents of absolute value up to 42440090988473300209087976590; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1497y^3 - 6sy^2 + 2485196y + 5144s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2994y^8 + 7211401y^6 + 8278771584y^4 + 4739146009936y^2 + 616016782981760$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34606612328200 = 2^3 \cdot 5^2 \cdot 11 \cdot 47 \cdot 2543 \cdot 131611$:

$$\begin{pmatrix} 1730330616410 & 665384873580 & 896179888394 & 1137553599693 & 1270429925962 & 336531032077 & 1119072287583 & 1727037198790 & 1448226960918 & 1678143938401 \\ 0 & 10 & 0 & 4 & 6 & 5 & 8 & 0 & 3 & 7 \\ 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 196 factors with signed exponents of absolute value up to 224982216132742352795028223699; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (4 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 274)y^3 + (32s + 112610)y^2 + (490s + 1782800)y + (8000s + 22827000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 224672y^7 + 26695866y^6 + 1593744920y^5 \\ & + 60266274140y^4 + 1516591629600y^3 + 25828669141000y^2 + 2639103 \\ & 65600000y + 2011018169000000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 1 \ -1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $11028989958892 = 2^2 \cdot 2757247489723$:

$$\begin{pmatrix} 5514494979446 & 2618256060158 & 3319464559781 & 1351164409893 & 4824009945099 & 2835652319873 & 5424819910396 & 5248600430587 & 3999676076568 & 815881425666 \\ 0 & 2 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 306 factors with signed exponents of absolute value up to 34053130; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 274)y^3 + (32s + 112610)y^2 + (490s + 1782800)y + (8000s + 22827000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 224672y^7 + 26695866y^6 + 1593744920y^5 \\ & + 60266274140y^4 + 1516591629600y^3 + 25828669141000y^2 + 2639103 \\ & 65600000y + 2011018169000000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 1 \ -1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $200469146154680 = 2^3 \cdot 5 \cdot 13 \cdot 385517588759$:

$$\begin{pmatrix} 100234573077340 & 44871537816188 & 23327344337645 & 86790156234051 & 32157992542773 & 54258146731631 & 12392539743258 & 78105122436465 & 57741376764562 & 50378541817906 \\ 0 & 2 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 291 factors with signed exponents of absolute value up to 49101263; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 274)y^3 + (32s + 112610)y^2 + (490s + 1782800)y + (8000s + 22827000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 549y^8 + 224672y^7 + 26695866y^6 + 1593744920y^5 + 60266274140y^4 + 1516591629600y^3 + 25828669141000y^2 + 263910365600000y + 2011018169000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 1 \ -1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $78673189270000 = 2^4 \cdot 5^4 \cdot 13^2 \cdot 46552183$:

$$\begin{pmatrix} 6051783790 & 4849650 & 3473044500 & 4549931290 & 981268594 & 6048767317 & 5118394763 & 269035266 & 5883573778 & 3151208575 \\ 0 & 130 & 70 & 10 & 115 & 85 & 125 & 80 & 99 & 54 \\ 0 & 0 & 10 & 0 & 1 & 1 & 4 & 5 & 0 & 8 \\ 0 & 0 & 0 & 10 & 4 & 5 & 0 & 5 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 292 factors with signed exponents of absolute value up to 80679732; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 2431y^3 + (-20s - 41720)y^2 + (560s - 586034)y + (37440s + 55718936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 4858y^8 - 73716y^7 + 4904573y^6 + 316624648y^5 + 13679143964y^4 - 743487973872y^3 - 31869757610684y^2 + 910906394568352y + 35737998355948096$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944}\right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $23328435046648 = 2^3 \cdot 29 \cdot 37 \cdot 2717664847$:

$$\begin{pmatrix} 5832108761662 & 5040872477075 & 4399817404695 & 3433385733532 & 2880739741715 & 3411967324992 & 4660159833363 & 2991839361562 & 258514776876 & 3990591178780 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 184 factors with signed exponents of absolute value up to 10690372358; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 2431y^3 + (-20s - 41720)y^2 + (560s - 586034)y + (37440s + 55718936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 4858y^8 - 73716y^7 + 4904573y^6 + 316624648y^5 + \\ & 13679143964y^4 - 743487973872y^3 - 31869757610684y^2 + 9109063945 \\ & 68352y + 35737998355948096 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right) s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34221031829920 = 2^5 \cdot 5 \cdot 13 \cdot 37 \cdot 4447 \cdot 99991$:

$$\begin{pmatrix} 2138814489370 & 1788347725198 & 724733819226 & 739031695364 & 1586823100520 & 1559220199762 & 2024767834750 & 1257104707363 & 1835324174168 & 399714773945 \\ 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 179 factors with signed exponents of absolute value up to 5330784735; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 2431y^3 + (-20s - 41720)y^2 + (560s - 586034)y + (37440s + 55718936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 4858y^8 - 73716y^7 + 4904573y^6 + 316624648y^5 + \\ & 13679143964y^4 - 743487973872y^3 - 31869757610684y^2 + 9109063945 \\ & 68352y + 35737998355948096 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $322813702177280 = 2^9 \cdot 5 \cdot 37 \cdot 389 \cdot 8761141$:

$$\begin{pmatrix} 5043964096520 & 517972930926 & 4515589265244 & 601299924262 & 558944926565 & 2101125304032 & 47259213896 & 576745777600 & 1063922116709 & 4463379852256 \\ 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 4 & 0 & 3 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 183 factors with signed exponents of absolute value up to 390369661; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1130y^3 - 20sy^2 - 1316000y + 8000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2260y^8 - 1355100y^6 + 6338004000y^4 - 5717875200000y^2 + 1489946240000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $16908048949088 = 2^5 \cdot 11^2 \cdot 59 \cdot 211 \cdot 350771$:

$$\begin{pmatrix} 96068459938 & 52202978938 & 31476526946 & 69117829246 & 43426832426 & 90147757002 & 37189300412 & 50445758877 & 86747348008 & 26830996644 \\ 0 & 22 & 21 & 6 & 13 & 18 & 18 & 19 & 10 & 16 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 310 factors with signed exponents of absolute value up to 496579981; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1130y^3 - 20sy^2 - 1316000y + 8000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2260y^8 - 1355100y^6 + 6338004000y^4 - 5717875200000y^2 + 148994624000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $267585311999500 = 2^2 \cdot 5^3 \cdot 11^2 \cdot 181 \cdot 24435899$:

$$\begin{pmatrix} 2432593745450 & 1569739141190 & 752875540384 & 1016810925366 & 28359707852 & 1169787635720 & 2039111369938 & 1600869988218 & 1699613782007 & 2193210113066 \\ 0 & 55 & 19 & 40 & 0 & 54 & 12 & 15 & 12 & 21 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 348 factors with signed exponents of absolute value up to 17422080028; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1130y^3 - 20sy^2 - 1316000y + 8000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2260y^8 - 1355100y^6 + 6338004000y^4 - 571787520000y^2 + 148994624000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $738812636320 = 2^5 \cdot 5 \cdot 11 \cdot 419779907$:

$$\begin{pmatrix} 92351579540 & 36730778152 & 1661939816 & 89598651484 & 80836332736 & 40356147804 & 88043681630 & 15037405572 & 18843641337 & 6603887778 \\ 0 & 4 & 3 & 1 & 3 & 2 & 2 & 2 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 343 factors with signed exponents of absolute value up to 26190595427; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.